Raja R. Sambasivan

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Tufts University
Computer Science Department
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Experience

Ankur & Mari Sahu Assistant Professor — *August 2019 – Present*
Tufts University, Computer Science Department

Research Scientist — *November 2016 – July 2019*
Boston University, Mass Open Cloud & Red Hat Collaboratory

Postdoctoral Researcher — *June 2013 – September 2016*
Carnegie Mellon University, eXpressive Internet Architecture (XIA) Group

Consultant — *November 2014 – December 2014*
Huawei, storage group

PhD Student — *June 2006 – May 2013*
Carnegie Mellon University, Parallel Data Lab

Software Engineering Intern — *May 2010 – December 2010*
Google

Research Intern — *July 2007 – March 2008*
HP Labs

Systems Programmer — *June 2004 – May 2006*
Carnegie Mellon University, Parallel Data Lab

Education

Ph.D., Electrical & Computer Engineering, May 2013
Carnegie Mellon University, Pittsburgh, PA
Advisor: Greg Ganger
Dissertation: Diagnosing performance changes in distributed systems by comparing request flows

M.S., Electrical & Computer Engineering, May 2004
Carnegie Mellon University, Pittsburgh, PA

B.S., Electrical & Computer Engineering w/minor in Computer Science, May 2003
Carnegie Mellon University, Pittsburgh, PA

Honours & Awards

NSF CAREER Award (2024-2029)

Nominee, Faculty using technology in innovative ways, May 2022

Nominee, Tufts senior survey significant impact, September 2021

Finalist, Facebook Distributed Systems Research RFP, March 2020. (The Worklfow motif: A powerful, widely-useful abstraction for diagnosing problems in distributed applications.)

Ankur & Mari Sahu Endowed Assistant Professorship, August 2019

Best poster, EMC University Day 2012 (*Diagnosing performance changes by comparing request flows*)

Best paper, SIGMETRICS 2007 (*Modeling the relative fitness of storage*)

Featured in Piled Higher & Deeper (PhDComics), February 14th, 2007 strip

Best paper, FAST 2005 (*Ursa Minor: versatile cluster-based storage*)

Teaching Experience

Course developer & instructor **Fall 2025, Fall 2024, Fall 2022, Fall 2021, Fall 2020, Spring 2020**

Tufts University COMP / CS 118, Cloud Computing

This class covers cloud-computing concepts and technologies. The curriculum includes lectures on research papers, homework assignments, and projects.

	Fall'25	Fall'24	Fall'22	Fall'21	Fall'20	Spring'20
Enrollment	35	40	48	42	27	37
Overall rating (max 5)	4.17	3.45	3.49	4.04	3.96	3.52
Instructor rating (max 5)	4.23	3.74	3.68	4.15	4.12	3.98

Course developer & instructor **Spring 2026, Spring 2025, Spring 2024, Spring 2023, Spring 2022, Spring 2021, Fall 2019**

Tufts University COMP / CS 110/150-DCC/151-DCC, Special Topics: Debugging Cloud Computing

This class introduces students to modern microservices-based distributed systems and state-of-the-art techniques for diagnosing problems in them. The curriculum emphasizes reading and discussing research papers on logging, distributed tracing, and network telemetry. Students perform an in-depth, semester-long, research project. Their final goal is to submit a research paper similar to those published in systems conferences.

	Spring'25	Spring'24	Spring'23	Spring'22	Spring'21	Fall'19
Enrollment	8	17	14	19	17	5
Overall rating (max 5)	4.88	3.86	4.00	4.07	4.52	4.35
Instructor rating (max 5)	4.88	3.93	4.00	4.40	4.46	4.71

Guest lecturer & project mentor **Spring 2017, Spring 2018 (guest lecturer only)**

Boston University CS 528, Cloud Computing

This class provides an overview of cloud-computing concepts via a curriculum that emphasizes reading research papers and semester-long projects. The class usually consists of 30-50 undergraduates and Master's students. I guest lectured on topics related to networking and problem diagnosis. In the Spring of 2017, I additionally mentored students on projects related to distributed tracing.

Course co-developer & co-instructor **Fall 2013**

Carnegie Mellon University CS 15-719, Advanced Cloud Computing

This class provides an overview of cloud-computing concepts. The curriculum emphasizes reading research papers, lectures, projects, and exams. The class consisted of about 30 Master's students and a few PhD students when I taught it.

Teaching Assistant **Fall 2005 & Spring 2010**

Carnegie Mellon University ECE 18-746, Storage Systems

This class covers a broad range of material, including hard-disk architecture, file-system design and debugging, RAID, and object-based storage. It usually consists of 40-60 Master's students and PhD students.

Professional service

Program committees:

- USENIX NSDI: 2026 (Spring & Fall), 2025 (Spring & Fall), 2023 (Spring & Fall)
- ACM SoCC: 2021
- ACM SIGCOMM: 2026
- ACM SIGMETRICS: 2022 (three cycles)
- IEEE ACSOS: 2020
- NENS: 2019

- IEEE Transactions on Services Computing: 2019, 2015
- IEEE Transactions on Software Engineering: 2017
- ACM HotStorage: 2014, 2013

External reviewing: SIGMETRICS 2021, SIGCOMM CCR 2018, EuroSys 2017

Panel member: NSF 2020, NSF Cloud Review 2017, NSF 2016

Session chair: USENIX NSDI 2025, 2023; ACM SoCC 2021; 2016 NENS 2019, 2017

Reading groups: Tufts systems (2020-present) BU diagnosis (2017-2019), CMU network diagnosis (2013-2014), CMU diagnosis (2011-2013)

Working groups: Red Hat telemetry co-lead (F'20 - S'22)

University service

Tufts CS Department:

- Computing committee, chair, F'25-S'26, F'24 - S'25,
- Systems faculty search committee: F'24 - S'25
- JEDI committee S'24
- Programming Languages faculty search committee: F'23 - S'24
- Ada Lovelace faculty search committee: F'22 - S'23
- Opportunity faculty search committee: S'22
- Graduate committee F'21 - S'22, F'20 - S'21
- Graduate student open-house co-organizer: F'21 - S'22, F'20 - S'21, F'19 - S'20
- Graduate admissions committee: F'20 - S'21, F'19 - S'20
- Colloquium co-organizer: F'25-S'26, F'24 - S'25, F'22 - S'23, F'21 - S'22, F'20 - S'21
- Wrote initial version of Tufts CS graduate student handbook supplement (Summer 2020)
- Software systems master's program development committee : F'19 - S'20

Tufts University:

- Faculty research support advisory committee (FRSAC) F'21 - S'26 (five-year term)

Prospectus, Proposal, & Dissertation committees

Noah Martin, Tufts University, Dissertation Committee Chair, Defended March 2026
Title: The Digital Divide in the Era of Artificial Intelligence - A Systems Perspective

Darby Huye, Tufts University, Dissertation Committee Chair, Defended February 2026
Title: Characterizing Microservices and Accommodating Tracing Observability Loss

Abdullah Bin Faisal, Tufts University, Dissertation Committee, Defended August 2025
Title: Leveraging Machine Learning for Improved Distributed System Performance

Hifza Khalid, Tufts University, Dissertation Committee, Defended September 2024
Title: Leveraging Machine Learning for Improved Distributed System Performance

Maziar Amiraski, Tufts University, Dissertation Committee, Defended April 2024
Title: Cost effective machine Learning For computer architecture design

Mohsin Bashir, Tufts University, Dissertation Committee, Defended December 2023
Title: Improving the performance of cloud applications: A multi-layered approach

Mania Abdi, Northeastern University, Proposal & Dissertation Committee, Defended September 2022
Title: Characterizing, debugging, and performance optimization of cloud applications using graph processing

Emre Ates, Boston University, Proposal & Dissertation Committee, Defended June 2020
Title: Automating telemetry- and trace-based analytics on large-scale distributed systems

Qualifying exam committees

Sarah Abowitz, Tufts University, December 2023

Tomislav Zabcic-Matic, Tufts University, May 2023 & December 2023

Darby Huye, Tufts University, March 2023

Carson Powers, Tufts University, December 2022

Max Liu, Tufts University, November 2022

Noah Martin, Tufts University, November 2022

Zhaoqi Zhang, Tufts University, April 2022

Hifza Khalid, Tufts University, December 2020

Abdullah Faisal, Tufts University, April 2020

Mohsin Bashir, Tufts University, April 2020

Mentoring & outreach

Mentor, USENIX NSDI PhD-student mentoring session: 2025, 2023
Mentor, MIT Primes high-school research program: Spring 2017-Spring 2025
Letter Writer, Letters to a Pre-Scientist middle-school outreach program (Fall 2018 - Spring 2019)
CS Grand Awards Judge, Intel Science & Engineering Fair Finals (2015, 2012)
Presenter, Carnegie Science Center Buhl Planetarium (2010)

Mentoring

Current PhD advisees (most recent to least recent):

- Tony Astolfi, part-tme PhD student, Tufts University (Spring 2025 -)
- Tomislav Zabcic-Matic, PhD student, Tufts University (Summer 2021 -)
- Darby Huye, MS & PhD student, Tufts University (MS: Fall 2020, PhD: Spring 2021 -)
- Max Liu, PhD student, Tufts University (Fall 2020 -)
- Zhaoqi Zhang, PhD student, Tufts University (Fall 2020 -)

Current Undergraduate advisees (most recent to least recent):

- Ha Nguyen (Summer 2025 -)

Previous students (most recent to least recent):

- Govind Velamoor, High-School Student'25 (Spring 2024 - Spring 2025)
  Next step: Yale University
- Adrita Samata, High-School Student'26 (Spring 2023 - Fall 2024)
  Next step: Continuing as high-school student
- Mona Ma, MS'23, MS Project Student (Fall'22 - Spring'24)
  Next step: SDE at Amazon
- Sarah Abowitz, Current PhD Student, PhD Student (Fall 2021 - Spring 2024)
  Next step: Changed advisor
- Henry Han, High-School Student'23 (Spring 2023)
  Next step: Continuing as high-school student
- Anshul Rastogi, High-School Student'23 (Spring 2021 - Spring 2023)
  Next step: University of Connecticut
- Joey Dong, High-School Student'23 (Spring 2022 - Spring 2023)
  Next step: MIT

- Alexander Ellis, MS'21, MS thesis student (Summer 2021 - Fall 2021)
  Thesis: Emplacing New Tracing: Adding OpenTelemetry to Envoy
  Next step: Continuing at Google
- Tanmay Gupta, Lexington High School'23 (Spring 2021 - Fall 2021)
  Next step: Continuing as high-school student
- Runze Si, MS'21, MS Project Student (Summer 2020 - Fall 2020)
  Next step: Unknown
- Neel Bhalla, Lexington High School'20 (Spring 2018 - Summer 2020)
  Next step: Northeastern University
- Lily Sturmann, MS'19, MS thesis student @ Harvard Extension School (Spring 2017 - Spring 2019)
  Thesis: Using Performance Variation for Instrumentation Placement in Distributed Systems
  Next step: Red Hat
- Harshal Sheth, Westford Academy' High School18 (Spring 2017 - Summer 2018)
  Next step: Yale University
- Andrew Sun, Westford Academy High School'18 (Spring 2017 - Summer 2018)
  Next step: Northeastern University

Other students I've worked with or mentored (most recent to least recent)

- Mania Abdi, PhD'23, NorthEastern University
  Advisor: Peter Desnoyers
- Mert Toslali, PhD'23, Boston University
  Advisor: Ayse Coskun
- Vishwanath Seshagiri, PhD student, Emory University
  Advisor: Avani Wildani
- Shihao Zhong, MS'21, Tufts University
- Emre Ates, PhD, Boston University (Fall 2018 - Spring 2020)
  Advisor: Ayse Coskun
- Emine Ugur Kaynar, PhD, Boston University
  Advisor: Orran Krieger
- Shwen (Jethro) Sun, MS'18, Boston University
  Advisor: Orran Krieger
- Da Yu, PhD, Brown University
  Advisor: Rodrigo Fonseca
- Golsana Ghaemi, PhD, Boston University
  Advisor: Orran Kreiger
- David Tran-Lam, PhD, University of Wisconsin-Madison
  Advisor: Aditya Akella
- Ilari Shafer, PhD, Carnegie Mellon University
  Advisor: Greg Ganger
- William Wang, MS, Carnegie Mellon University
  Advisor: Greg Ganger

Mentee awards & achievements

2023: Joey Dong and Anshul Rastogi, high-school students, presented at the AMS-PME student poster session at the Joint Mathematics Meeting.

2022: Tanmay Gupta and Anshul Rastogi, high-school students, presented at the AMS-PME poster session at the Joint Mathematics Meeting.

2020: Neel Bhalla, high-school student, awarded 2nd prize at the MA State Science Fair.

2018: Harshal Sheth and Andrew Sun, high-school students, presented at the Red Hat Developers' Conference Devconf.us.

2017: Harshal Sheth and Andrew Sun, high-school students, named Siemens Competition Semi-Finalists.

Advisee internships

Summer 2023: Sarah Abowitz, DynaTrace, Mentor: Stefan Achleithner.

Summer 2022 - Spring 2023: Darby Huye, Meta, Mentor: Yuri Shkuro

Funding

CAREER: Principled yet practical observability for a microservices-based cloud. *PI*: Raja R. Sambasivan $609,452. NSF Award # 2340128. July 2024 - June 2029.

Automated instrumentation frameworks for diagnosing inadvertent causes of resource contention. *PI*: Raja R. Sambasivan. $15,000. Indonesian Ministry of Research and Education. Awarded May 2023.

A powerful, widely-useful abstraction for diagnosing performance problems in distributed applications. *PI*: Raja R. Sambasivan. $60,000. Red Hat. Awarded January 2022.

CSR: Small: A just-in-time, instrumentation framework for diagnosing performance problems in distributed applications. *PI*: Raja R. Sambasivan, *Co-PIs*: Ayse K. Coskun, Orran Krieger. $460,249. NSF Award # 1815323. Re-issued as NSF Award # 2016178. October 2018 - September 2022.

Interpreting publications

In my field, the faculty member leading a publication is listed last. The lead student author is listed first. Academic conference publications are highly valued. I have annotated my student advisees in the publication list below with asterisks (*) and students I mentored, but did not advise, with daggers (†). Individuals who presented publications at academic or industrial venues are additionally annotated with deltas (Δ). I generally (but not always) define mentorship relationships as collaborations with other faculty members' students during my time as a professor or those with junior students during my time as a Ph.D student, postdoc, or research scientist.

Peer-reviewed publications

Dynamic read & write optimization with TurtleKV Tony Astolfi*, Vidya Silai, Darby Huye*, Lan Liu*, Raja R. Sambasivan, Johes Bater. Accepted to the 52nd International Conference on Very Large Databases (VLDB'26). August 31th to September 4th, 2026.

Systemizing and mitigating topological inconsistencies in Alibaba's microservice call-graph datasets. Darby Huye*, Lan Liu*, Raja R. Sambasivan. In Proceedings of the 15th International Conference on Performance Engineering (ICPE'24). May 7th to May 11th, 2024. Kensington, England. DOI: https://doi.org/10.1145/3629526.3645043

Lifting the veil on Meta's microservice architecture: Analyses of topology and request workflows. Darby Huye*Δ, Yuri Shkuro, Raja R. Sambasivan. In Proceedings of the 2023 USENIX Annual Technical Conference (ATC'23). July 10th to July 12th, 2023. Boston, MA, USA. ISBN: 978-1-939133-35-9.

VAIF: Variance-based Automated Instrumentation Framework. Mert Toslali†, Emre Ates†, Darby Huye*, Zhaoqi Zhang*, Lan Liu*, Samantha Puterman, Ayse K. Coskun, Raja R. Sambasivan. Operating Systems Review. 56(1), 42-50, 2022. DOI: https://doi.org/10.1145/3544497.3544504.

[SoK] Identifying mismatches between microservice testbeds and industrial perceptions of microservices. Vishwanath Seshagiri†, Darby Huye*, Lan Liu*, Avani Wildani, Raja R. Sambasivan. Journal of Systems Research, 2(1), 2022. DOI: https://doi.org/10.5070/SR32157839.

Automating instrumentation choices for performance problems in distributed applications with VAIF. Mert Toslali†Δ, Emre Ates†, Alexander Ellis*, Zhaoqi Zhang*, Darby Huye*, Lan Liu*, Samantha Puterman, Ayse K. Coskun, Raja R. Sambasivan. In Proceedings of the 12th ACM Symposium on Cloud Computing (SoCC'21). November 1st to November 3rd, 2021. Seattle, WA, USA. DOI: https://doi.org/10.1145/3472883.3487000.

D3N: A multi-layer cache for the rest of us. Emine Ugur Kaynar†Δ, Mania Abdi†, Mohammad Hossein Hajkazemi, Ata Turk, Raja R. Sambasivan, Larry Rudolph, Peter Desnoyers, Orran Krieger. In proceedings of the 2019 IEEE International Conference on Big Data (BigData'19). December 9th to December 12th, 2019. Los Angeles, CA, USA.

An automated, cross-layer instrumentation framework for diagnosing performance problems in distributed applications. Emre Ates†, Mert Toslali†Δ, Richard Megginson, Orran Krieger, Ayse K. Coskun, Raja R. Sambasivan. In Proceedings of the 10th ACM Symposium on Cloud Computing (SoCC'19). November 20th to November 23rd, 2019. Santa Cruz, CA.

Bootstrapping evolvability for inter-domain routing with D-BGP. Raja R. SambasivanΔ, David Tran-Lam†, Aditya Akella, Peter Steenkiste. In proceedings of the ACM 2017 SIGCOMM Conference (SIGCOMM'17). August 21th to August 25th, 2017. Los Angeles, CA, USA.

Principled workflow-centric tracing of distributed systems. Raja R. SambasivanΔ, Ilari Shafer†, Jonathan Mace, Rodrigo Fonseca, Gregory R. Ganger. In proceedings of the 7th ACM Symposium on Cloud Computing (SoCC'16). October 5th to October 7th, 2016. Santa Clara, CA, USA.

Bootstrapping evolvability for inter-domain routing. Raja R. SambasivanΔ, David Tran-Lam†, Aditya Akella, Peter Steenkiste. In proceedings of the 14th ACM Workshop on Hot Topics in Networks (HotNets'15). November 16th to November 17th, 2015. Philadelphia, PA, USA.

Visualizing request-flow comparison to aid performance diagnosis in distributed systems. Raja R. SambasivanΔ, Ilari Shafer†, Michelle L. Mazurek, Gregory R. Ganger. IEEE Transactions on Visualization and Computer Graphics 19(12), December 2013. In proceedings of Information Visualization 2013. Atlanta, Georgia, USA.

Specialized storage for big numeric time series. Ilari ShaferΔ, Raja R. Sambasivan, Anthony Rowe, Gregory R. Ganger. In proceedings of the 5th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage'13). June 27th to June 28th, 2013. San Jose, CA, USA.

Automated diagnosis without predictability is a recipe for failure. Raja R. SambasivanΔ, Gregory R. Ganger. In proceedings of the 4th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud'12). June 12th to June 13th, 2012. Boston, MA, USA.

Diagnosing performance changes by comparing request flows. Raja R. SambasivanΔ, Alice X. Zheng, Michael De Rosa, Elie Krevat, Spencer Whitman†, Michael Stroucken, William Wang†, Lianghong Xu, Gregory R. Ganger. In proceedings of the 8th USENIX Symposium on Network Systems Design and Implementation (NSDI'11). March 30th to April 1st, 2011. Boston, MA, USA.

A transparently-scalable metadata service for the Ursa Minor storage system. Shafeeq SinnamohideenΔ, Raja R. Sambasivan, Likun Liu, James Hendricks, Gregory R. Ganger. In proceedings of the 2010 USENIX Annual Technical Conference (USENIX ATC'10). June 23rd to 25th, 2010. Boston, MA, USA.

Relative fitness modeling. Michael Mesnier, Matthew Wachs, Raja R. Sambasivan, Alice Zheng, Raja R. Sambasivan, Gregory R. Ganger. Research Highlights, Communications of the ACM. April 2009.

Categorizing and differencing system behaviours. Raja R. SambasivanΔ, Alice X. Zheng, Eno Thereska, Gregory R. Ganger. Appears in the proceedings of the 2nd International Workshop on Hot Topics in Autonomic Computing (HotAC II). June 15th, 2007. Jacksonville, Florida, USA.

Modeling the relative fitness of storage. Michael MesnierΔ, Matthew Wachs, Raja R. Sambasivan, Alice X. Zheng, Gregory R. Ganger. In proceedings of the International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS'07). June 12th to 16th, 2007. San Diego, CA, USA.

//TRACE: **parallel trace replay with approximate causal events**. Michael MesnierΔ, Matthew Wachs, Raja R. Sambasivan, Julio Lopez, James Hendricks, Gregory R. Ganger. In proceedings of the 5th conference on File and Storage Technologies (FAST'07). February 13th to 16th, 2007. San Jose, CA, USA.

Early experiences on the journey towards self-* storage. Michael Abd-El-Malek, William V. Courtright II, Chuck Cranor, Gregory R. Ganger, James Hendricks, Andrew J. Klosterman, Michael Mesnier, Manish Prasad, Brandon Salmon, Raja R. Sambasivan, Shafeeq Sinnamohideen, John D. Strunk, Eno Thereska, Matthew Wachs, Jay J. Wylie. In the Bulletin of the IEEE Computer Society Technical Committee on Data Engineering 29(3). Special issue on self-managing database systems. September 2006.

Ursa Minor: versatile cluster-based storage. Michael Abd-El-Malek, William V. Courtright II, Chuck Cranor, Gregory R. Ganger, James Hendricks, Andrew J. Klosterman, Michael Mesnier, Manish Prasad, Brandon Salmon, Raja R. Sambasivan, Shafeeq Sinnamohideen, John D. Strunk∆, Eno Thereska, Matthew Wachs, Jay J. Wylie. In the proceedings of the 4th USENIX conference on File and Storage Technologies (FAST'05). December 13th to 16th, 2005. San Francisco, CA, USA.

Replication policies for layered clustering of NFS servers. Raja R. Sambasivan∆, Andrew J. Klosterman, Gregory R. Ganger. Appears in the proceedings of the 13th Annual Meeting of the IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS'05). September 27th to 29th, 2005. Atlanta, Georgia, USA.

Technical Reports

Dynamic read & write optimization with TurtleKV. Tony Astolfi*, Vidya Silai, Darby Huye*, Lan Liu*, Raja R. Sambasivan, Johes Bater. arXiv:2509.10714 [cs.DB]. September 12th, 2025. Last revised December 8$^{\text{th}}$, 2025.

A widely-useful abstraction for distributed applications. Mania† Abdi, Peter Desnoyers, Mark Crovella, Raja R. Sambasivan. arXiv:2506.00749 [cs.DC]. May 30th, 2025. (Originally prepared in June 2020.)

Bootstrapping evolvability for inter-domain routing with D-BGP. Raja R. Sambasivan, David Tran-Lam†, Aditya Akella, Peter Steenkiste. Carnegie Mellon Computer Science Technical Report CMU-CS-16-117. June 2016.

So, you want to trace your distributed system? Key design insights from years of practical experience. Raja R. Sambasivan, Rodrigo Fonseca, Ilari Shafer†, Gregory R. Ganger. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-14-102. April 2014.

Visualizing request-flow comparison to aid performance diagnosis in distributed systems. Raja R. Sambasivan, Ilari Shafer†, Michelle L. Mazurek. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-13-104. May 2013. Supersedes CMU-PDL-12-102.

Visualizing request-flow comparison to aid performance diagnosis in distributed systems. Raja R. Sambasivan, Ilari Shafer†, Michelle L. Mazurek. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-12-102. May 2012.

Automation without predictability is a recipe for failure. Raja R. Sambasivan, Gregory R. Ganger. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-11-101. January 2011.

Diagnosing performance changes by comparing system behaviours. Raja R. Sambasivan, Alice X. Zheng, Elie Krevat, Spencer Whitman†, Michael Stroucken, William Wang†, Lianghong Xu, Gregory R. Ganger. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-10-107. July 2010. Supersedes CMU-PDL-10-103.

A transparently-scalable metadata service for the Ursa Minor storage system. Shafeeq Sinnamohideen, Raja R. Sambasivan, James Hendricks, Likun Liu, Gregory R. Ganger. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-10-102. March 2010.

Diagnosing performance problems by visualizing and comparing system behaviours. Raja R. Sambasivan, Alice X. Zheng, Elie Krevat, Spencer Whitman†, Gregory R. Ganger. Carnegie Mellon University Parallel Data Lab Technical Report CMU-PDL-10-103. February 2010.

Eliminating cross-server operations in scalable file systems. James Hendricks, Shafeeq Sinnamohideen, Raja R. Sambasivan, Gregory R. Ganger. Carnegie Mellon University Parallel Data Lab Technical Report CMU-PDL-06-105. May 2006.

Improving small file performance in object-based storage. James Hendricks, Raja R. Sambasivan, Shafeeq Sinnamohideen, Gregory R. Ganger. Carnegie Mellon University Parallel Data Lab Technical Report CMU-PDL-06-104. May 2006.

Selected project reports, Spring 2005 Advanced OS & Distributed Systems (15-712). Garth A. Gibson and Hyang-Ah Kim, Editors. Jangwoo Kim, Eriko Nurvitadhi, Eric Chung; Alex Nizhner, Andrew Biggadike, Jad Chamcham; Srinath Sridhar, Jeffrey Stylos, Noam Zeilberger; Gregg Economou, Raja R. Sambasivan, Terrence Wong; Elaine Shi, Yong Lu, Matt Reid; Amber Palekar, Rahul Iyer. Carnegie Mellon Computer Science Technical Report CMU-CS-05-138. May 2005.

Ursa Minor: Versatile cluster-based storage. Michael Abd-El-Malek, William V. Courtright II, Chuck Cranor, Gregory R. Ganger, James Hendricks, Andrew J. Klosterman, Michael Mesnier, Manish Prasad, Brandon Salmon, Raja R. Sambasivan, Shafeeq Sinnamohideen, John D. Strunk, Eno Thereska, Matthew Wachs, Jay J. Wylie. Carnegie Mellon University Parallel Data Laboratory Technical Report CMU-PDL-05-104. April 2005.

Patents

Managing execution of database queries. Stefan Kompress, Harumi Anne Kuno, Umeshwar Dayal, Janet Wiener, Raja Sambasivan. U.S. Patent 9,910,892. March 6^{th}, 2018.

Posters & WiP talks

Dynamic read & write optimization with TurtleKV. Tony Astolfi[*], Vidya Silai, Darby Huye[*], Lan Liu[*], Raja R. Sambasivan, Johes Bater. Presented at the 2026 North East Database Day (NEDB'26). January 17^{th}, 2026.

TurtleKV: A dynamically optimized key/value engine. Tony Astolfi[*Δ], Vidya Sila, Darby Huye[*], Lan Liu[*] Raja R. Sambasivan, Johes Bater. Three minute presentation and Poster at Tufts School of Engineering Graduate Research Spotlight. September 12^{th}, 2025. Boston, MA, USA.

Towards holes as a first-class primitive in distributed tracing. Darby Huye[*Δ], Tony Astolfi[*], Raja R. Sambasivan. In the poster session of the 19^{th} USENIX Symposium on Operating Systems Design & Implementation (SOSP'25). July 7^{th} to July 9^{th}, 2025. Boston, MA, USA.

TurtleKV: A dynamically optimized dey/value engine. Tony Astolfi[*Δ], Vidya Sila, Darby Huye[*], Lan Liu[*] Raja R. Sambasivan, Johes Bater. In the poster session of the 19^{th} USENIX Symposium on Operating Systems Design & Implementation (OSDI'25). July 7^{th} July 9^{th}, 2025. Boston, MA, USA.

TurtleDB: Highly tunable, high-performance key/value storage. Anthony Astolfi[*Δ], Vidya Sila, Raja R. Sambasivan, Johes Bater. In the poster session of the 2025 North East Database Day (NEDB'25). January 9^{th}, 2025. Boston, MA, USA.

Diagnosing behavioral causes of contention in distributed systems. Zhaoqi Zhang[*Δ], Tomislav Zabcic-Matic[*Δ], Mert Toslali[†], Ayse K. Coskun, Raja R. Sambasivan. In the poster session of the 20^{th} USENIX Symposium on Network Systems Design and Implementation (NSDI'23). April 17^{th} to April 19^{th}, 2023. Boston, MA, USA.

PADLocal: Privacy-aware debugging for microservices. Sarah Abowitz[*Δ], Daniel Votipka, Raja R. Sambasivan. In the poster session of the 20^{th} USENIX Symposium on Network Systems Design and Implementation (NSDI'23). April 17^{th} to April 19^{th}, 2023. Boston, MA, USA.

An automated, cross-layer instrumentation framework for diagnosing performance problems in distributed applications. Emre Ates[†], Mert Toslali[†Δ], Richard Megginson, Orran Krieger, Ayse K. Coskun, Raja R. Sambasivan. In the poster session of the 10^{th} ACM Symposium on Cloud Computing (SoCC'19). November 20^{th} to

November 23^{rd}, 2019. Santa Cruz, CA, USA.

Extending Workflow-centric tracing into the Linux Kernel. Golsana Ghaemi†∆, Orran Krieger, Raja R. Sambasivan. WiP talk at New England Networking and Systems Day 2019 (NENS'19). April 16^{th}, 2019. Boston, MA, USA.

LeitMotif: An abstraction for debugging distributed applications. Mania Abdi†∆, Golsana Ghaemi†, Ata Turk, Peter Desnoyers, Mark Crovella, Raja R. Sambasivan. Poster at New England Networking and Systems Day'19 (NENS'19). April 16^{th}, 2019. Boston, MA, USA.

An automated, cross-layer instrumentation framework for diagnosing performance problems in distributed applications. Emre Ates†∆, Mert Toslali†, Richard Megginson, Orran Krieger, Ayse K. Coskun, Raja R. Sambasivan. April 16^{th}, 2019. In the poster session of the 2019 New England Systems Day (NESD'19). Boston, MA, USA.

Pythia: A just-in-time instrumentation framework for distributed applications. Lily Sturmann†∆, Shwen (Jethro) Sun†, Orran Krieger, Ayse K. Coskun, Peter Portante, Raja R. Sambasivan. Poster at New England Networking and Systems Day'17 (NENS'17). December 7^{th}, 2017.

Generalizing request-flow comparison to more systems. Raja R. Sambasivan∆, William Wang†, Gregory R. Ganger. WiP talk at 23^{rd} ACM Symposium on Operating Systems Principles (SOSP'11). October 23^{rd} to October 26^{th}, 2011.

Software releases

Data and code accompanying *Systematizing and mitigating topological inconsistencies in Alibaba's microservice call-graph datasets* paper published in ICPE'24. https://doi.org/10.7910/DVN/SS9SIY. Additional code updates maintained at https://github.com/docc-lab/casper. Originally released March 28th, 2024.

Raw data for graphs published in *Lifting the veil on Meta's microservice architecture: Analyses of topology and request workflows* paper published in ATC'23. https://github.com/facebookresearch/distributed_traces. Originally released on July 10^{th}, 2023.

Code for VAIF prototype described in *Automating instrumentation choices for performance problems in distributed applications with VAIF* paper published in SoCC'21. https://github.com/docc-lab/pythia. Originally released February 6^{th}, 2023.

Code for Spectroscope prototype described in *Diagnosing performance changes by comparing request flows* paper published NSDI'11. https://github.com/RS1999ent/spectroscope.git. Originally released May 13^{th}, 2014.

Interviews & panels

Academia, industry, and debugging research. On-stage Interview at KubeCon Observability Summit, November 2019.

Industry conference talks

LeitMotif: An abstraction for debugging distributed applications. Mania Abdi†∆, Peter Desnoyers, Mark Crovella, Raja R. Sambasivan. Observability Practitioners Summit at KubeCon'19. November 19*th*, 2019. San Diego, CA, USA.

An automated cross-layer instrumentation framework for diagnosing performance problems in distributed applications. Emre Ates†∆, Lily Sturmann, Mert Toslali, Orran Krieger, Ayse K.Coskun, Raja R. Sambasivan. Observability Practitioners Summit at KubeCon'19. November 19*th*, 2019. San Diego, CA, USA.

Logging what matters: the Pythia just-in-time instrumentation framework. Lily Sturmann*∆, Emre Ates†, Mert Toslali†, Orran Krieger, Ayse K. Coskun, Raja R. Sambasivan. Observability Practitioners Summit at KubeCon'18. December 10^{th}, 2018. Seattle, WA, USA.

Invited Talks (I present)

Toward principled yet practical observability for microservices. Presented at the University of Toronto (May 18^{th}, 2025), UMass Amherst (April 18^{th}, 2025), and MathWorks (December 16^{th}, 2025).

Understanding and efficiently observing microservices. Presented at The University of Chicago. November 8^{th}, 2023.

Cross-job prefetching for object storage. Presented at the Mass Open Cloud & iScale IUCRC Workshop, Boston University. March 21^{st}, 2023.

Finding frequent patterns in distributed traces. Discussion at LightStep, November 2019.

Workflow-centric tracing and advanced diagnosis tools for the cloud ecosystem. Presented at Red Hat (June 2019), UMass Amherst (February 2019), George Washington University (March 2019), Tufts University (March 2019).

Toward a diagnosis plane for cloud computing. Presented at LightStep (April 2018), Columbia University (March 2018), Facebook (February 2018), Brown University (February 2018).

Diagnosis and inter-domain support for an Internet of clouds. Presented at Tufts University (October 2017), Yale University (March 2017), AT&T Labs (May 2016), Intel Labs (April 2016), NYU (April 2016).

Diagnosing performance changes by comparing request flows. Presented at UCSD (April 2014), Brown University (April 2012), NetApp RTP (September 2011), Google NYC (June 2011).

Guest lectures

Networking at Google: B4 & Jupiter Rising. Guest lecture in BU CS 528 (March 2018, March 2017).

When the cloud fizzles: Outages and how to debug them. Guest lecture in NU CS 6620 (April 2018) and BU CS 528 (April 2017).

A case study of the AWS outage on April 21^{st}, 2011. Guest lecture in CMU 15-719 (Fall 2015, Fall 2014).

Diagnosis via monitoring & tracing. Guest lecture in CMU 15-719 (Fall 2015, Fall 2014).

Mentions in books

Distributed tracing in practice: instrumenting, analyzing, and debugging microservices. Austin Parker, Daniel Spoonhower, Jonathan Mace, Rebecca Isaacs. O'Reilly. 2020. *VAIF work is discussed in Chapter 13*.

Mastering distributed tracing. Yuri Shkuro. Packt, 2019. *Mentioned in Foreward*.